

Internship Key Points

Parisara-Cycle - Green Commuter Guide Android Application

Project context: Parisara-Cycle is a Kotlin Android application that supports bicycle commuters with route planning, danger-zone reporting, pit-stop discovery, CO2 savings tracking, and optional nearby rider location sharing.

Project(s) Completed

- Completed the Parisara-Cycle Android app structure with login, map, stats, and profile modules.
- Implemented cycling-focused features such as route generation, danger-zone reporting, pit-stop markers, and buddy rider visibility.
- Prepared the project to work in local demo mode, with optional Firebase and Google Maps/Directions integration for future deployment.

Outputs/Results Achieved

- Generated a working app flow where users can sign in, select destinations, create routes, and view route distance.
- Enabled eco-stat tracking that calculates daily and monthly CO2 savings using distance travelled by cycle.
- Produced setup and user documentation to support installation, demonstration, and future project extension.

Application of Engineering Knowledge

- Applied Android architecture concepts by separating UI screens, ViewModels, repositories, data models, and utility classes.
- Used location-based engineering logic for current location, route distance calculation, nearby marker display, and rider filtering.
- Applied sustainable mobility calculations by converting cycled distance into estimated CO2 savings.

Tools/Techniques Used

- Used Kotlin, Jetpack Compose, Material 3, Gradle, and Android Studio project structure for app development.
- Used Firebase-ready repositories for Authentication, Firestore danger-zone storage, and Realtime Database buddy sharing.
- Used local fallback techniques such as SharedPreferences, mock pit-stops, local route generation, and keyless map rendering.

Impact of Your Work

- Improved cyclist safety by allowing users to report potholes, dangerous intersections, and blocked paths.
- Encouraged eco-friendly commuting by making CO2 savings visible through daily and monthly statistics.
- Made the project easier to demonstrate because it can run without mandatory Firebase setup or paid API keys.

Practical Implementation

- Built practical screens for authentication, route planning, eco-stat viewing, and profile-based live location sharing.
- Connected user actions such as tapping the map, long-pressing danger zones, toggling filters, and generating routes to ViewModel logic.
- Stored user sessions, reported danger zones, and eco-stat data locally, while keeping the design ready for cloud synchronization.